



Number: _____

Name: _____

Teacher: _____

GOSFORD HIGH SCHOOL

Preliminary HSC

Extension 1 Mathematics 2012

Assessment Task 2

TIME ALLOWED 80 minutes plus 5 minutes reading time

- Show all working for questions worth more than 1 mark.
- An approved calculator may be used

Marks

	Multiple choice	/5
	Question 1 Polynomials	/10
	Question 2 Ratio	/10
	Question 3 Polynomials	/10
	Question 4 Circle Geometry	/11
	Question 5 Polynomials	/10
Total		/56



Name: _____

Teacher: _____

Multiple-choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely, using a black pen.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct
↓

Start
here →

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐

Section 1 Multiple Choice (5 Marks)

1. Which of the following is a factor of $x^6 + 6x^2 - 4x - 3$
- A. $x + 1$ B. $x - 1$ C. $x + 3$ D. $x - 3$
2. Which of the following is a **NOT** a polynomial
- A. $\frac{5}{x^2} + 3x$ B. $(x - 2)(x - 3)$ C. 3^{-1} D. x^{16}
3. Which of the following is a monic polynomial of degree 4?
- A. $1 - x + x^2 - x^4$ B. $1 + 3x + 2x^2 + 4x^3$
C. $5x + x^4$ D. $4x$
4. If Q divides AB externally in the ratio 1:3, then A divides BQ in the ratio
- A. 3 : 1 B. -1 : 3 C. 1 : 2 D. 2 : 1
5. If α, β , and γ are the roots of $3x^3 - 5x + 4 = 0$, which of the following is true
- (i) $\alpha + \beta + \gamma = \frac{5}{3}$
(ii) $\alpha\beta\gamma = \frac{-4}{3}$
- A. (i) only B. (ii) only C. both (i) and (ii) D. neither (i) nor (ii)

Question 1 (10 Marks)	Polynomials	Start a New sheet of Paper	Marks
a) Find the remainder when $F(x) = x^3 + 2x^2 - 5x + 3$ is divided by $x + 2$.			1
b) Subtract $x^3 - 5x^2 + 3x$ from $x^4 - 2x^3 + 1$.			1
c) Sketch the graph of $P(x) = x(x - 3)(x + 1)$ labelling all intercepts.			2
(at least $\frac{1}{3}$ page sketch and rule your axes!)			
d) (i) Divide $P(x) = x^3 + 4x^2 + 1$ by $A(x) = x - 3$			2
(ii) Express your answer in the form $P(x) = A(x).Q(x) + R(x)$			1
e) $P(x) = x^3 + ax^2 + bx - 18$. Find a and b if $(x + 2)$ is a factor of $P(x)$ and			
-24 is the remainder when $P(x)$ is divided by $(x - 1)$			3
Question 2 (10 Marks) Ratio Formula		Start a New sheet of Paper	
a) A is the point $(-3, 4)$ and B is the point $(2, -1)$. Find the coordinates of			
(i) the point P which divides AB internally in the ratio 5: 3.			3
(ii) the point Q which divides BA externally in the ratio 1: 2			3
b) (i) For the same points A and B above, write down an expression for P (x, y) in			1
terms of k , which divides AB in the ratio $k: 1$. Answer in the form $x =$, and $y =$.			
(ii) The line $2x - 3y + 4 = 0$ cuts AB at the point D. Using the result from (i),			
find in what ratio D divides AB?			3

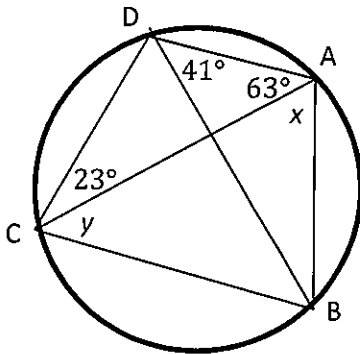
Question 3 (10 Marks) Polynomials**Start a New sheet of Paper****Marks**

- a) Solve $3^{2x} - 4(3^x) + 3 = 0$ 3
- b) Solve $(3x^2 + 2x)^2 - 9(3x^2 + 2x) + 8 = 0$ 3
- c) Sketch the graph of $y = (x + 1)^2(1 - x)^3(x + 2)$ 4

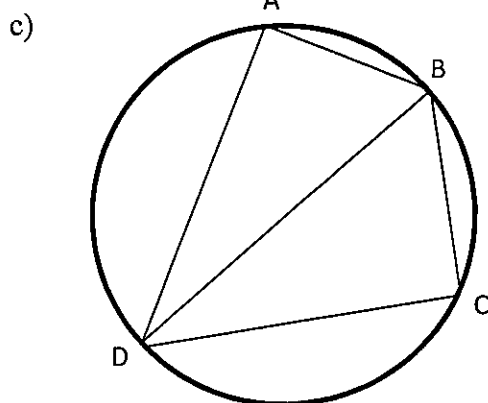
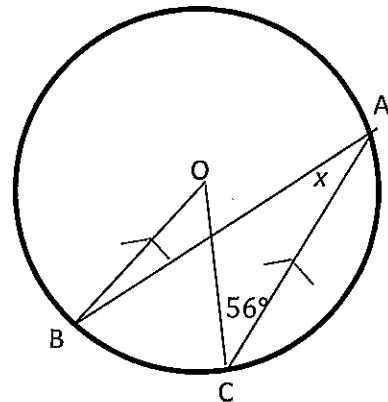
labelling all x and y intercepts (at least $\frac{1}{3}$ page sketch and rule your axes!)

Question 4 (11 Marks) Circle Geometry.**Start a New sheet of Paper****Copy or trace each diagram onto your writing pad.**

- a) Find the values of y and x giving all reasons 4



- b) O is the centre of this circle.
Find the value of x giving all reasons. 2



A, B, C and D lie on a circle. BD bisects $\angle ABC$ and $\angle ADC$.

Letting $\angle DBC = x$ and $\angle BDC = y$

(i) Prove $\triangle ABC$ is isosceles 2

(ii) Prove BD is a diameter 3

You must give all reasons.

Hint: Join AC

Question 5 (10 Marks) Polynomials**Start a New sheet of Paper**

- a) If α, β , and γ are the roots of $x^3 - x^2 + 4x - 5 = 0$ form the equation with roots $2\alpha, 2\beta$, and 2γ . 3
- b) $P(x)$ leaves a remainder of -6 when divided by $x - 1$ and 4 when divided by $x + 1$. Find the remainder when $P(x)$ is divided by $(x^2 - 1)$. 2
- c) Two of the roots of $2x^4 - 5x^3 + cx^2 + 5x - 2 = 0$ are reciprocals of each other, while the remaining two roots are opposites of each other.
- (i) Find the sum of the roots 1
- (ii) Find the product of the roots. 1
- (iii) Hence find the four roots and the value of c . 3

Solutions to 2012 Extension 1 Preliminary HSC Assessment Task 2.

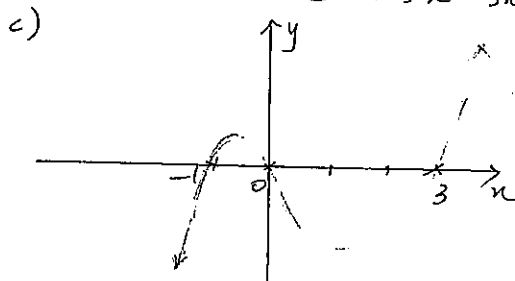
Multiple choice

1. B 2. A 3. C 4. D 5. B

Question 1

a) $F(-2) = -8 + 8 + 10 + 3 = 13$

b) $x^4 - 2x^3 + 1 - x^3 + 5x^2 - 3x$
 $= x^4 - 3x^3 + 5x^2 - 3x + 1$



d) i)

$$\begin{array}{r} x^2 + 7x + 21 \\ x-3 \overline{) x^3 + 4x^2 + 0x + 1} \\ \underline{x^3 - 3x^2} \\ 7x^2 \\ \underline{7x^2 - 21x} \\ 21x + 1 \\ \underline{21x - 63} \\ 64 \end{array}$$

d) ii) $x^3 + 4x^2 + 1 = (x-3)(x^2 + 7x + 21) + 64$

e) $P(2) = 0 \Rightarrow -8 + 4a - 2b - 18 = 0$

$\text{or } 2a - b = 13 \quad \text{--- (1)}$

$P(1) = -24 \Rightarrow 1 + a + b - 18 = -24$

$\text{or } a + b = -7 \quad \text{--- (2)}$

$\textcircled{1} + \textcircled{2}$

$3a = 6$

$a = 2$

Sub into $\textcircled{2} \quad 2 + b = -7$

$b = -9$

$\therefore a = 2, \quad b = -9$

Question 2

A $(-3, 4)$ B $(2, -1)$ $k:l = 5:3$

$x = \frac{kx_1 + lx_2}{k+l}$

$y = \frac{ky_1 + ly_2}{k+l}$

$x = \frac{10 - 9}{8}$

$y = \frac{-5 + 12}{8}$

P is $(\frac{1}{8}, \frac{7}{8})$

Question 2 (continued)

B $(2, -1)$ A $(-3, 4)$ $k:l = -1:2$

$x = \frac{-1 \times 3 + 2 \times 2}{-1 + 2}$

$y = \frac{-1 \times 4 + 2 \times -1}{-1 + 2}$

$x = 7$

$y = -6$

Q is $(7, -6)$

b) i) $x = \frac{2k-3}{k+1}$ $y = \frac{-k+4}{k+1}$

ii) $2x - 3y + 4 = 0$

$2\left(\frac{2k-3}{k+1}\right) - 3\left(\frac{-k+4}{k+1}\right) = -4$

$4k - 6 + 3k - 12 = -4k - 4$

$11k = 14$

$k = \frac{14}{11}$

$\therefore$ D divides AB internally in ratio 14:11.

Question 3

a) Put $u = 3^x$ $u^2 - 4u + 3 = 0$
 $(u-3)(u-1) = 0$
 $u = 3 \text{ or } 1$

$\therefore 3^x = 3$

$\text{or } 3^x = 1$

$x = 1$

$\text{or } x = 0$

b) Put $u = 3x^2 + 2x$

$\therefore u^2 - 9u + 8 = 0$

$(u-8)(u-1) = 0$

$u = 8 \text{ or } 1$

$3x^2 + 2x - 8 = 0$

$\text{or } 3x^2 + 2x - 1 = 0$

$(3x-4)(x+2) = 0$

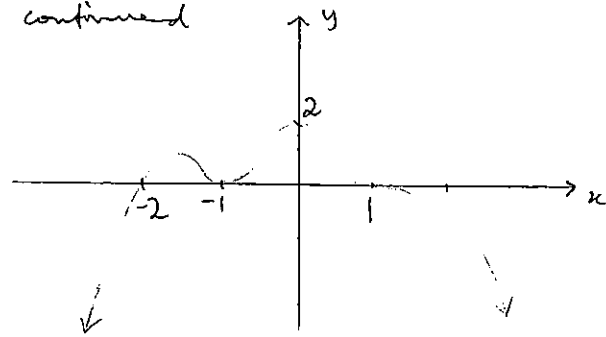
$3x = 4$
 $x = \frac{4}{3}$

$(3x-1)(x+1) = 0$

$x = \frac{1}{3} \text{ or } -1$

$x = \frac{4}{3} \text{ or } -2$

$\therefore x = \frac{4}{3}, 2, \frac{1}{3} \text{ or } -1$

Question 3 continuedQuestion 4 a) $y = 41^\circ$ (Angles at circumference standing

on same arc AB are equal

$$\angle CDB = 180 - (21 + 41 + 63) = 53^\circ \text{ (angle sum of } \triangle = 180^\circ)$$

 $x = 53^\circ$ (Angles in same segment standing on arc BC are equal.)

b) $\angle BDC = 56^\circ$ (Alternate $\angle$'s are equal $BO \parallel CA$)
 $\angle BAC = x = \frac{1}{2} \times 56^\circ = 28^\circ$ (Angle at centre is twice $\angle$ at circumference standing on same arc BC)

c) i) $\angle ADB = \angle BDC = y$ (BD bisects $\angle ADC$ (given))
 $\angle ACB = \angle ADB = y$ (Angles at circumference standing on arc AB are equal.)
 $\angle CAB = \angle CDB = y$ (Angles at circumference standing on arc BC are equal)

$$\therefore \angle CAB = \angle ACB = y$$

$$\therefore AB = BC \text{ (opposite equal angles)}$$

 $\therefore \triangle ABC$ is isosceles.

ii) $\angle DAC = \angle DBC = x$ (angles at circumference standing on arc DC are equal)

$$\angle ABD = \angle DBC = x \text{ (DB bisects } \angle ABC \text{ (given))}$$

 $\therefore$ In $\triangle ABD$

$$\angle BDA + \angle DAB + \angle ABD = 180^\circ \text{ (angle sum of } \triangle ABD = 180^\circ)$$

$$\therefore y + (x+y) + x = 180^\circ$$

$$2x + 2y = 180^\circ$$

$$\therefore x + y = 90^\circ$$

But Angle in a semi-circle is a right angle

 $\therefore BD$ is a diameter
Question 5

a) $x^3 - x^2 + 4x - 5 = 0$

$$2 + \beta + \gamma = -\frac{b}{a} = 1$$

$$2\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = 4$$

$$2\beta\gamma = -\frac{d}{a} = 5$$

$$\therefore 2\alpha + 2\beta + 2\gamma = 2 \times 1 = 2$$

$$2\alpha \cdot 2\beta + 2\beta \cdot 2\gamma + 2\gamma \cdot 2\alpha = 4 \times 4 = 16$$

$$2\alpha \cdot 2\beta \cdot 2\gamma = 8 \times 5 = 40.$$

$$\therefore \text{Equation is } x^3 - 2x^2 + 16x - 40 = 0$$

b) Let remainder be $ax + b$
 $\therefore P(x) = (x+1)(x-1)Q(x) + ax + b$

$$P(1) = -6 \therefore -6 = a + b \quad \text{--- (1)}$$

$$P(-1) = 4 \therefore 4 = -a + b \quad \text{--- (2)}$$

$$\text{(1) + (2)} \quad 2b = -2$$

$$b = -1$$

$$\text{(1) - (2)} \quad 2a = -10$$

$$\text{and } a = -5$$

c) i) Let roots be $\alpha, \frac{1}{\alpha}, \beta, -\beta$

$$\therefore \text{Sum of roots} = \alpha + \frac{1}{\alpha} = \frac{5}{2} \quad \text{--- (1)}$$

$$\text{ii) Product of roots} = -\beta^2 = -\frac{2}{2}$$

$$\beta^2 = 1$$

$$\text{--- (2)}$$

iii) Solving (1) $2\alpha^2 - 5\alpha + 2 = 0$

$$(2\alpha - 1)(\alpha - 2) = 0$$

$$\alpha = \frac{1}{2} \text{ or } 2$$

$$\begin{array}{r} 2x \times 1 \\ x \times -2 \end{array}$$

Solving (2) $\beta = \pm 1$

$$\therefore \text{The four roots are } \underline{2, \frac{1}{2}, +1, -1}$$

Sub any root to find c

eg Sub $x = 1$

$$2 - 5 + c + 5 - 2 = 0$$

$$\underline{c = 0}$$